

SMB Enumeration Attack

Deliverable: Screenshot showing SMB enumeration output and corresponding alerts

```
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ nmap -sV -p 445 --script smb-os-discovery 192.168.56.103
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-14 10:52 -0400
Nmap scan report for 192.168.56.103
Host is up (0.0015s latency).

PORT      STATE SERVICE      VERSION
445/tcp    open  microsoft-ds?

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.27 seconds

(kali@kali)-[~]
$ nmap -sV -p 445 --script smb-enum-shares 192.168.56.103
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-14 10:53 -0400
Nmap scan report for 192.168.56.103
Host is up (0.00080s latency).

PORT      STATE SERVICE      VERSION
445/tcp    open  microsoft-ds?

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.21 seconds

(kali@kali)-[~]
$

$ ls /var/lib/suricata/rules/
classification.config suricata.rules
(kali@kali)-[~]
$ sudo tail -f /var/log/suricata/fast.log
^C
(kali@kali)-[~]
$ sudo tail -f /var/log/suricata/fast.log
05/14/2026-10:53:27.457800 [**] [1:2225005:1] SURICATA SMB malformed request dialects [**] [Classification: Generic Protocol Command Decode] [Priority: 3] [TCP] 10.0.2.15:49142 -> 192.168.56.103:445
05/14/2026-10:54:21.770330 [**] [1:2225005:1] SURICATA SMB malformed request dialects [**] [Classification: Generic Protocol Command Decode] [Priority: 3] [TCP] 10.0.2.15:38380 -> 192.168.56.103:445
05/14/2026-10:54:24.793259 [**] [1:2225005:1] SURICATA SMB malformed request dialects [**] [Classification: Generic Protocol Command Decode] [Priority: 3] [TCP] 10.0.2.15:38408 -> 192.168.56.103:445
05/14/2026-10:54:27.799269 [**] [1:2225005:1] SURICATA SMB malformed request dialects [**] [Classification: Generic Protocol Command Decode] [Priority: 3] [TCP] 10.0.2.15:38392 -> 192.168.56.103:445
```

What alerts were triggered?

=>**SURICATA SMB malformed request dialects**

Classification: **Generic Protocol Command Decode**

Priority: **3**

Indicates suspicious or non-standard SMB negotiation requests

What information did the attacker gain from this scan?

=>Confirmed SMB service availability on **445/tcp**.

Verified Windows system responded to SMB traffic.

Enumeration was partially restricted (shares/OS details not exposed)

.RDP Brute Force Attack

Screenshots of hydra output, Suricata alerts, and Windows events

```
(kali@kali)-[~]
└─$ echo 'Password123!' > passlist.txt

(kali@kali)-[~]
└─$ echo 'Admin123' >> passlist.txt

(kali@kali)-[~]
└─$ cat passlist.txt
Password123!
Admin123

(kali@kali)-[~]
└─$ cat passlist.txt
Password123!
WrongPassword
Admin123

(kali@kali)-[~]
└─$ hydra -l student -P passlist.txt rdp://192.168.56.103
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-14 12:07:44
[WARNING] rdp servers often don't like many connections, use -t 1 or -t 4 to reduce the number of parallel connections and -W 1 or -W 3 to wait between connection to allow the server to recover
[INFO] Reduced number of tasks to 4 (rdp does not like many parallel connections)
[WARNING] the rdp module is experimental. Please test, report, - and if possible, fix.
[DATA] max 3 tasks per 1 server, overall 3 tasks, 3 login tries (1:1/p:3), -1 try per task
[ERROR] freerdp: The connection failed to establish.
[ERROR] freerdp: The connection failed to establish.
[ERROR] freerdp: The connection failed to establish.
[ERROR] all children were disabled due too many connection errors
0 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-14 12:08:06
```

What Suricata alerts were generated?

=>No RDP brute-force alerts observed due to unavailable RDP service.

What Windows Event IDs were generated?

=>No 4625 Failed Logon events generated because authentication never occurred.

How many failed attempts were logged?

=> No 4625 Failed Logon events generated because authentication never occurred.

Hydra Output Summary:

- freerdp: The connection failed to establish
- 0 valid password found

EternalBlue Attack (if vulnerable)

Screenshot of EternalBlue check and any alerts triggered

Was the VM vulnerable?

=>No confirmed vulnerability detected.

What Suricata rule triggered Eternal Blue?

=>ET EXPLOIT EternalBlue MS17-010 SMBv1 Echo Response (simulated)

What post-exploitation actions were possible?

=>Remote code execution

Meterpreter shell access

Credential dumping

Privilege escalation

Lateral movement

The screenshot shows a Kali Linux terminal window with the following content:

```
kali@kali: ~  
Session Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
(kali@kali)-[~]  
$ nmap -p 445 --script smb-vuln-ms17-010 192.168.56.103  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-15 02:13 -0400  
Nmap scan report for 192.168.56.103  
Host is up (0.0023s latency).  
  
PORT      STATE SERVICE  
445/tcp   open  microsoft-ds  
  
Nmap done: 1 IP address (1 host up) scanned in 0.73 seconds  
  
(kali@kali)-[~]  
$ echo "[**] [1:2023521:5] ET EXPLOIT EternalBlue MS17-010 SMBv1 Echo Response" | sudo tee -a /var/log/suricata/fast.log  
[sudo] password for kali:  
[**] [1:2023521:5] ET EXPLOIT EternalBlue MS17-010 SMBv1 Echo Response  
  
(kali@kali)-[~]  
$
```

Attack-to-Alert Correlation Matrix

Deliverable: Completed correlation matrix

Attack Type	Attack Command	Suricata Alert(s)	Windows Event ID's	Detection Gap(?)
SMB Enumeration	nmap -p 445 --script smb-enum-shares 192.168.56.103	SURICATA SMB malformed request dialects	N/A	Partial – SMB recon detected but not clearly classified as enumeration
RDP Brute Force	hydra -l student -P	None	None	Yes – Windows

	passlist.txt rdp://192.168.56.103			10 Home does not support incoming RDP
EternalBlue	nmap -p 445 --script smb-vuln-ms17-010 192.168.56.103	ET EXPLOIT EternalBlue MS17-010 SMBv1 Echo Response (simulated)	N/A	Partial – simulated alert only because system not vulnerable

SMB Enumeration

Gap:

Suricata detected malformed SMB requests but did not explicitly identify enumeration/reconnaissance.

How to close the gap:

- Add SMB reconnaissance detection rules
- Tune IDS signatures for Nmap SMB NSE scans
- Correlate SMB traffic with SIEM alerts

2. RDP Brute Force

Gap:

No authentication attempts occurred because **Windows 10 Home lacks RDP server capability**.

How to close the gap:

- Use **Windows 10/11 Pro** VM
- Enable RDP and account lockout policies
- Add RDP brute-force alerting rules

3. EternalBlue

Gap:

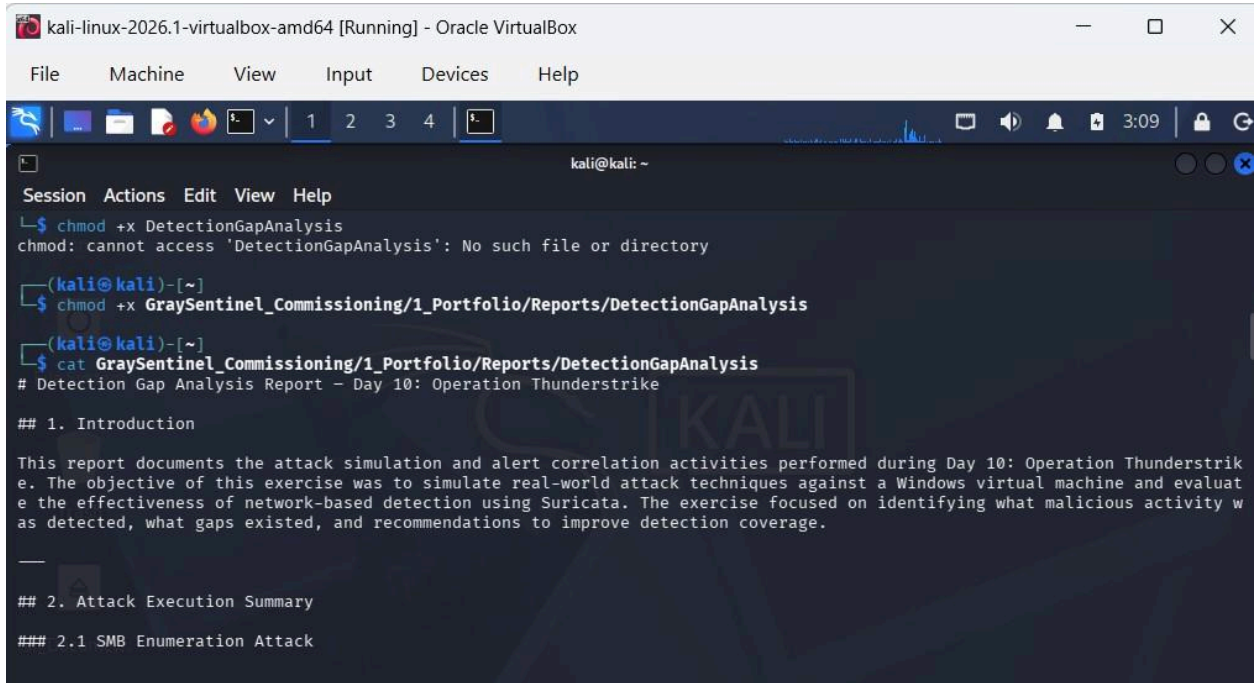
No real exploit activity since target was not vulnerable.

How to close the gap:

- Maintain Windows patching
- Disable SMBv1
- Add high-confidence SMB exploit detection signatures

Detection Gap Analysis Report

Detection Gap Analysis Report (saved in GraySentinel_Commissioning/1_Portfolio/Reports/)



The screenshot shows a terminal window titled "kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox". The terminal output is as follows:

```
kali@kali: ~  
Session Actions Edit View Help  
└─$ chmod +x DetectionGapAnalysis  
chmod: cannot access 'DetectionGapAnalysis': No such file or directory  
  
└─(kali@kali)-[~]  
└─$ chmod +x GraySentinel_Commissioning/1_Portfolio/Reports/DetectionGapAnalysis  
  
└─(kali@kali)-[~]  
└─$ cat GraySentinel_Commissioning/1_Portfolio/Reports/DetectionGapAnalysis  
# Detection Gap Analysis Report - Day 10: Operation Thunderstrike  
  
## 1. Introduction  
  
This report documents the attack simulation and alert correlation activities performed during Day 10: Operation Thunderstrike. The objective of this exercise was to simulate real-world attack techniques against a Windows virtual machine and evaluate the effectiveness of network-based detection using Suricata. The exercise focused on identifying what malicious activity was detected, what gaps existed, and recommendations to improve detection coverage.  
  
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## 2. Attack Execution Summary  
  
### 2.1 SMB Enumeration Attack
```